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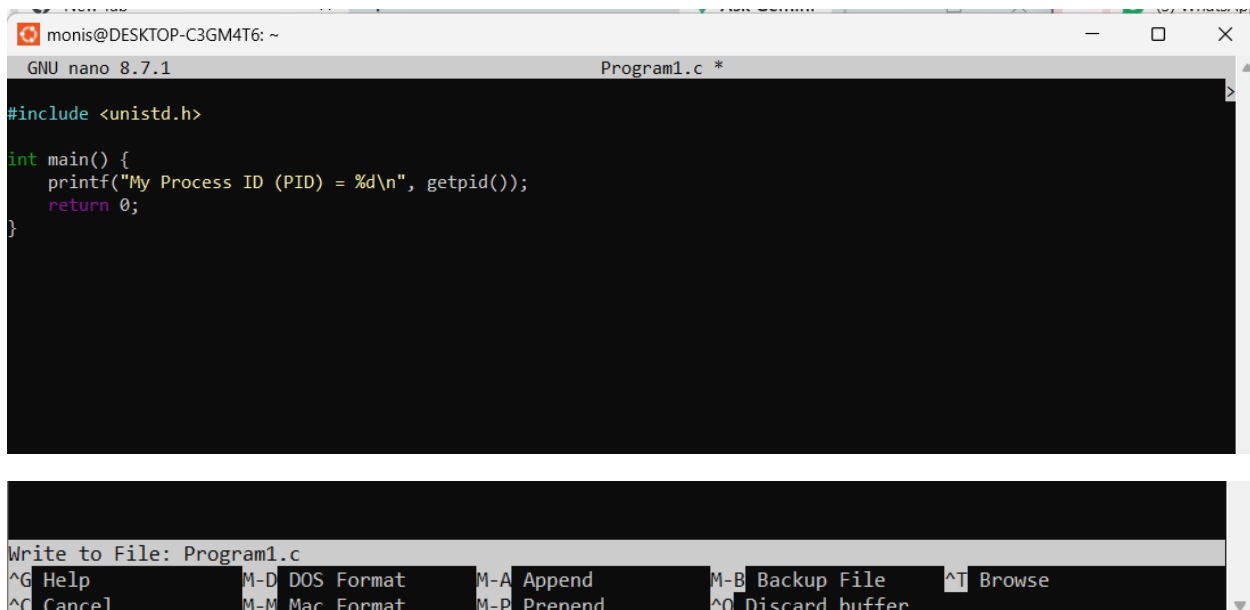
Date:14-8-26

Roll No: 2520030537

Branch: B.Tech CSE

Subject: OSSP

Program-1

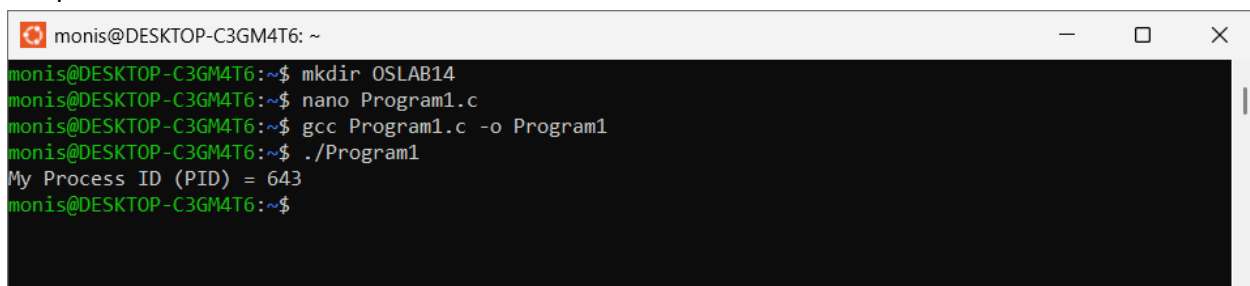


```
monis@DESKTOP-C3GM4T6: ~  
GNU nano 8.7.1 Program1.c *  
  
#include <unistd.h>  
  
int main() {  
    printf("My Process ID (PID) = %d\n", getpid());  
    return 0;  
}
```

Write to File: Program1.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:



```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ mkdir OSLAB14  
monis@DESKTOP-C3GM4T6:~$ nano Program1.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program1.c -o Program1  
monis@DESKTOP-C3GM4T6:~$ ./Program1  
My Process ID (PID) = 643  
monis@DESKTOP-C3GM4T6:~$
```

Program-2

```
Select monis@DESKTOP-C3GM4T6: ~
GNU nano 8.7.1                                Program2.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("My Process ID (PID) = %d\n", getpid());
    printf("My Parent Process ID (PPID) = %d\n", getppid());
    sleep(30); // keeps process alive for 30 seconds
    return 0;
}
```

Write to File: Program2.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:

```
monis@DESKTOP-C3GM4T6: ~
monis@DESKTOP-C3GM4T6:~$ nano Program2.c
monis@DESKTOP-C3GM4T6:~$ gcc Program2.c -o Program2
monis@DESKTOP-C3GM4T6:~$ ./Program2
My Process ID (PID) = 730
My Parent Process ID (PPID) = 530
```

Sleep(30);

```
monis@DESKTOP-C3GM4T6: ~
monis@DESKTOP-C3GM4T6:~$ ./Program2
My Process ID (PID) = 818
My Parent Process ID (PPID) = 530
monis@DESKTOP-C3GM4T6:~$
```

Program-3

```
monis@DESKTOP-C3GM4T6: ~  
GNU nano 8.7.1 Program3.c *  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {  
    pid_t pid = fork();  
  
    if (pid == 0) {  
        // Child process  
        printf("Child Process\n");  
        printf("PID = %d\n", getpid());  
    } else {  
        // Parent process  
        printf("Parent Process\n");  
        printf("PID = %d\n", getpid());  
    }  
  
    return 0;  
}
```

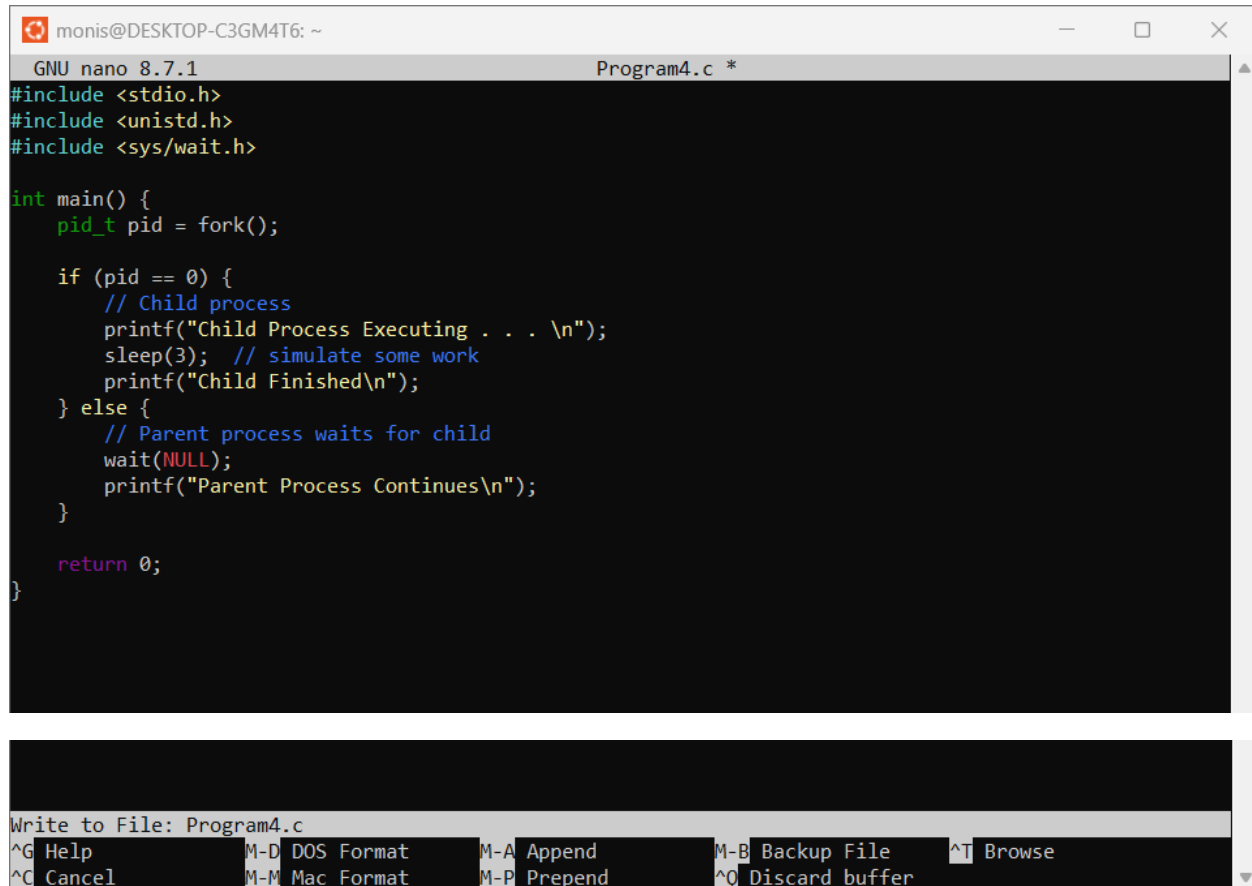
Write to File: Program3.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ nano Program3.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program3.c -o Program3  
monis@DESKTOP-C3GM4T6:~$ ./Program3  
Parent Process  
PID = 783  
Child Process  
PID = 784
```

Program-4

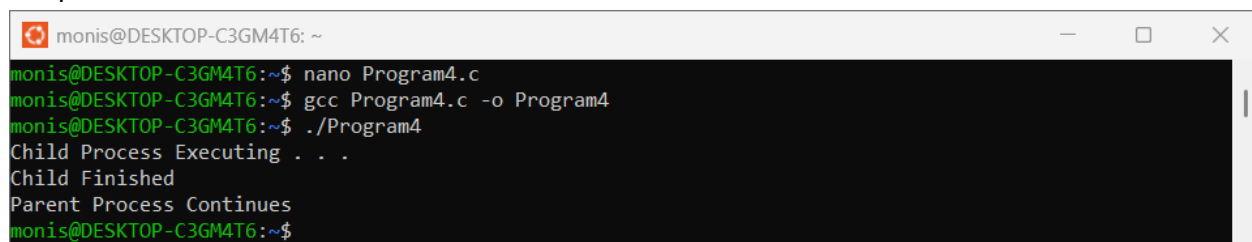


```
monis@DESKTOP-C3GM4T6: ~  
GNU nano 8.7.1 Program4.c *  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main() {  
    pid_t pid = fork();  
  
    if (pid == 0) {  
        // Child process  
        printf("Child Process Executing . . . \n");  
        sleep(3); // simulate some work  
        printf("Child Finished\n");  
    } else {  
        // Parent process waits for child  
        wait(NULL);  
        printf("Parent Process Continues\n");  
    }  
  
    return 0;  
}
```

Write to File: Program4.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:



```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ nano Program4.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program4.c -o Program4  
monis@DESKTOP-C3GM4T6:~$ ./Program4  
Child Process Executing . . .  
Child Finished  
Parent Process Continues  
monis@DESKTOP-C3GM4T6:~$
```

Program4.1->Multiple Child Processes

```
monis@DESKTOP-C3GM4T6: ~  
GNU nano 8.7.1 Program4.1.c *  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {  
    for (int i = 1; i <= 3; i++) {  
        pid_t pid = fork();  
  
        if (pid == 0) {  
            // Child process  
            printf("Child %d PID = %d\n", i, getpid());  
            return 0; // child exits after printing  
        }  
    }  
    return 0;  
}
```

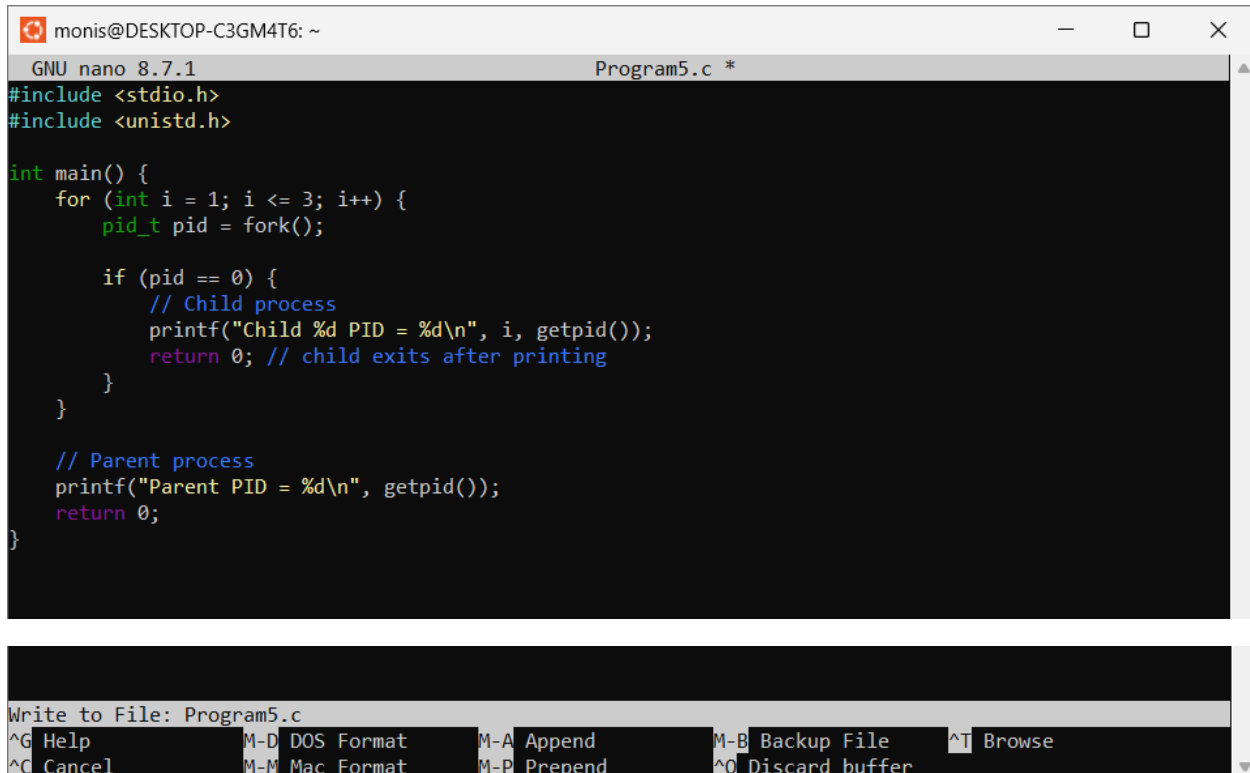
Write to File: Program4.1.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ nano Program4.1.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program4.1.c -o Program4.1  
monis@DESKTOP-C3GM4T6:~$ ./Program4.1  
Child 1 PID = 1165  
Child 2 PID = 1166  
Child 3 PID = 1167  
monis@DESKTOP-C3GM4T6:~$
```

Program-5



```
monis@DESKTOP-C3GM4T6: ~
GNU nano 8.7.1 Program5.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    for (int i = 1; i <= 3; i++) {
        pid_t pid = fork();

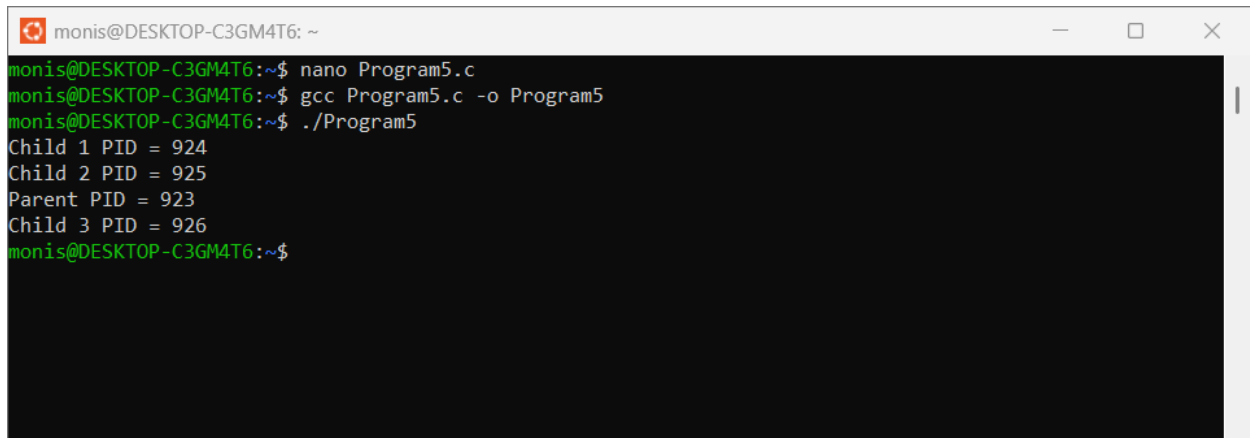
        if (pid == 0) {
            // Child process
            printf("Child %d PID = %d\n", i, getpid());
            return 0; // child exits after printing
        }

        // Parent process
        printf("Parent PID = %d\n", getpid());
        return 0;
    }
}
```

Write to File: Program5.c

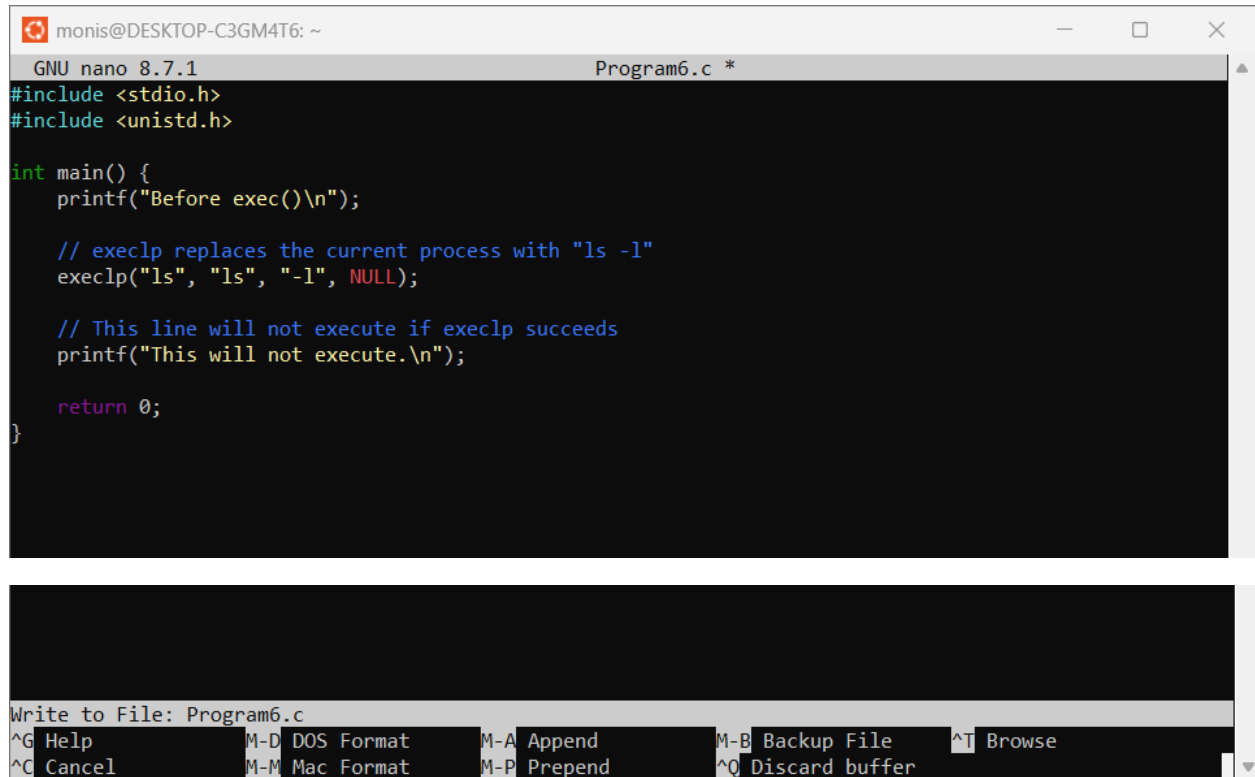
^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:



```
monis@DESKTOP-C3GM4T6: ~$ nano Program5.c
monis@DESKTOP-C3GM4T6: ~$ gcc Program5.c -o Program5
monis@DESKTOP-C3GM4T6: ~$ ./Program5
Child 1 PID = 924
Child 2 PID = 925
Parent PID = 923
Child 3 PID = 926
monis@DESKTOP-C3GM4T6: ~$
```

Program-6



```
monis@DESKTOP-C3GM4T6: ~
GNU nano 8.7.1 Program6.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Before exec()\n");

    // execlp replaces the current process with "ls -l"
    execlp("ls", "ls", "-l", NULL);

    // This line will not execute if execlp succeeds
    printf("This will not execute.\n");

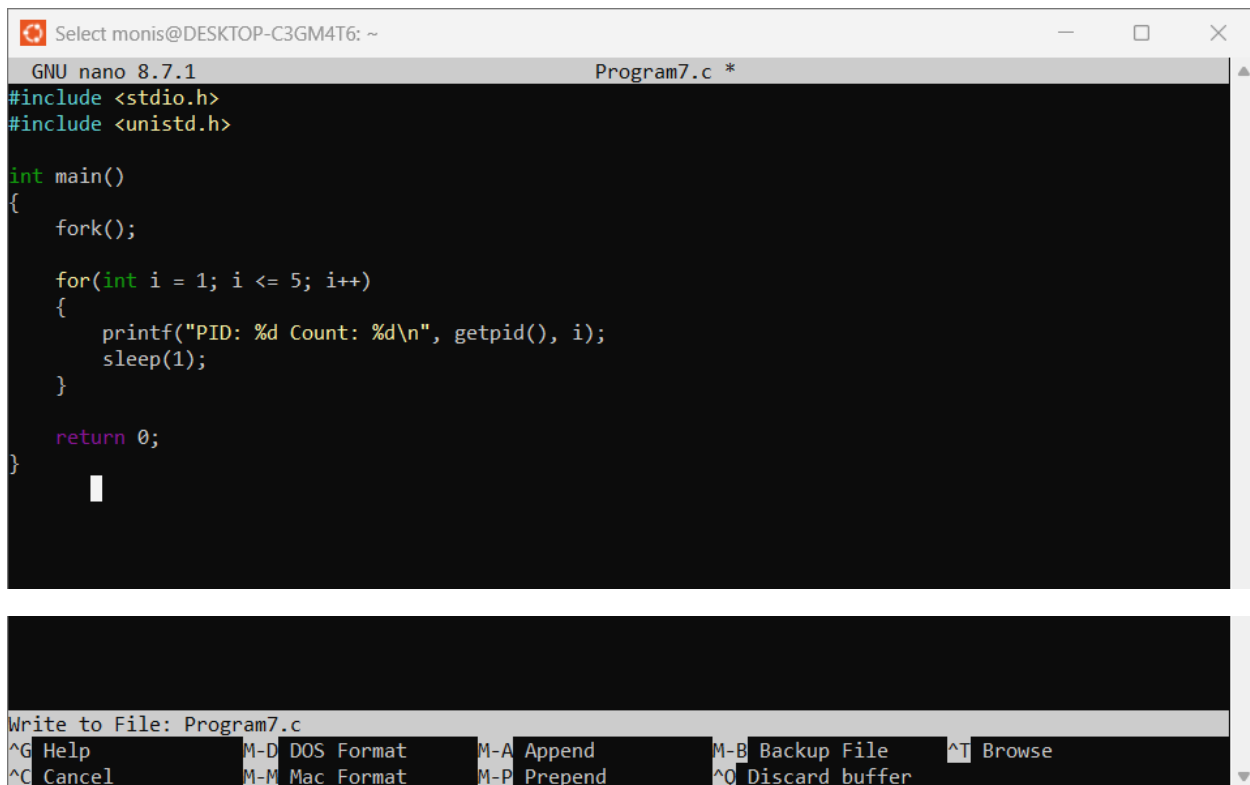
    return 0;
}

Write to File: Program6.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

Output:

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ nano Program6.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program6.c -o Program6  
monis@DESKTOP-C3GM4T6:~$ ./Program6  
Before exec()  
total 276  
drwxr-xr-x 2 monis monis 4096 Jul 31 14:21 Linux-System-Programming-Week1  
drwxr-xr-x 3 monis monis 4096 Aug 5 03:23 OSLAB  
drwxr-xr-x 2 monis monis 4096 Aug 14 14:37 OSLAB14  
drwxr-xr-x 2 monis monis 4096 Aug 7 12:08 OSSP-Practical  
-rwxr-xr-x 1 monis monis 16000 Aug 14 14:38 Program1  
-rw-r--r-- 1 monis monis 353 Aug 14 14:38 Program1.c  
-rw-r--r-- 1 monis monis 353 Aug 14 14:36 Program1.c.save  
-rwxr-xr-x 1 monis monis 16088 Aug 14 14:48 Program2  
-rw-r--r-- 1 monis monis 235 Aug 14 14:45 Program2.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 14:47 Program3  
-rw-r--r-- 1 monis monis 336 Aug 14 14:47 Program3.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 14:52 Program4  
-rw-r--r-- 1 monis monis 418 Aug 14 14:51 Program4.c  
-rwxr-xr-x 1 monis monis 16040 Aug 14 14:54 Program5  
-rw-r--r-- 1 monis monis 376 Aug 14 14:53 Program5.c  
-rwxr-xr-x 1 monis monis 16000 Aug 14 14:56 Program6  
-rw-r--r-- 1 monis monis 288 Aug 14 14:55 Program6.c  
drwxr-xr-x 2 monis monis 4096 Aug 4 10:08 Simple-Commands-Week1  
-rw-r--r-- 1 monis monis 0 Jul 31 03:59 example.txt  
drwxr-xr-x 2 monis monis 4096 Jul 31 04:00 foldername  
-rwxr-xr-x 1 monis monis 15952 Aug 4 03:53 hello  
-rw-r--r-- 1 monis monis 75 Aug 4 03:51 hello.c  
drwxr-xr-x 2 monis monis 4096 Jul 28 04:00 moni  
drwxr-xr-x 2 monis monis 4096 Jul 28 04:09 monis  
-rwxr-xr-x 1 monis monis 16080 Aug 14 09:52 program2  
-rw-r--r-- 1 monis monis 229 Aug 14 09:51 program2.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 10:06 program3  
-rw-r--r-- 1 monis monis 260 Aug 14 10:06 program3.c  
-rwxr-xr-x 1 monis monis 16040 Aug 14 10:13 program4  
-rw-r--r-- 1 monis monis 163 Aug 14 10:13 program4.c  
-rwxr-xr-x 1 monis monis 16000 Aug 14 10:17 program5  
-rw-r--r-- 1 monis monis 155 Aug 14 10:17 program5.c  
-rwxr-xr-x 1 monis monis 15952 Jul 31 09:36 sample1  
-rw-r--r-- 1 monis monis 77 Jul 31 09:18 sample1.c  
-rw----- 1 monis monis 0 Jul 31 09:17 sample1.c.save
```


Program-7



```
GNU nano 8.7.1 Program7.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

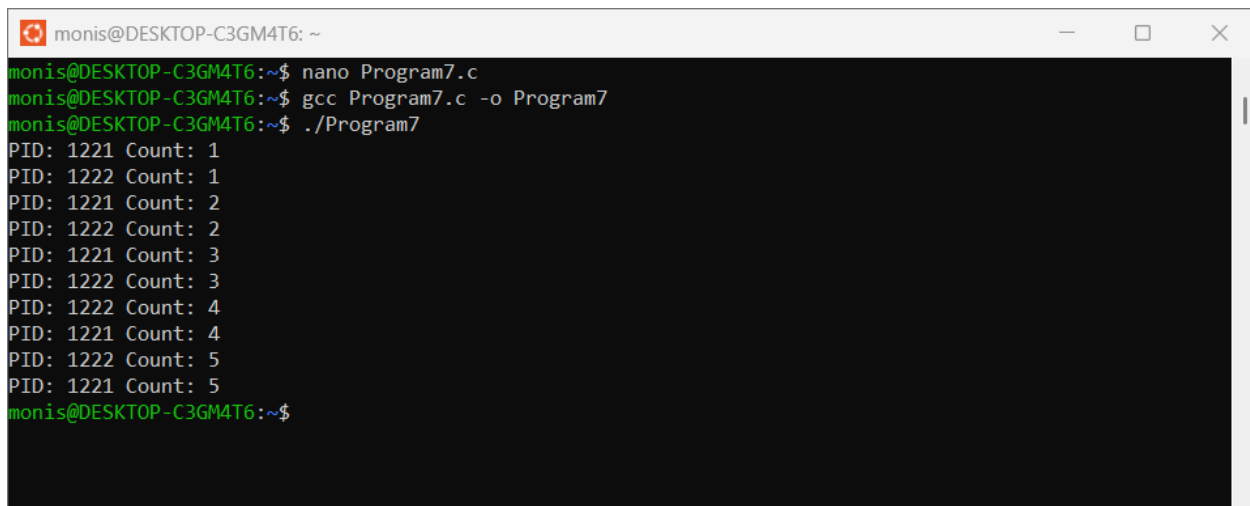
    for(int i = 1; i <= 5; i++)
    {
        printf("PID: %d Count: %d\n", getpid(), i);
        sleep(1);
    }

    return 0;
}
```

Write to File: Program7.c

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

Output:



```
monis@DESKTOP-C3GM4T6: ~$ nano Program7.c
monis@DESKTOP-C3GM4T6: ~$ gcc Program7.c -o Program7
monis@DESKTOP-C3GM4T6: ~$ ./Program7
PID: 1221 Count: 1
PID: 1222 Count: 1
PID: 1221 Count: 2
PID: 1222 Count: 2
PID: 1221 Count: 3
PID: 1222 Count: 3
PID: 1222 Count: 4
PID: 1221 Count: 4
PID: 1222 Count: 5
PID: 1221 Count: 5
monis@DESKTOP-C3GM4T6: ~$
```

Output:

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ nano Program6.c  
monis@DESKTOP-C3GM4T6:~$ gcc Program6.c -o Program6  
monis@DESKTOP-C3GM4T6:~$ ./Program6  
Before exec()  
total 276  
drwxr-xr-x 2 monis monis 4096 Jul 31 14:21 Linux-System-Programming-Week1  
drwxr-xr-x 3 monis monis 4096 Aug 5 03:23 OSLAB  
drwxr-xr-x 2 monis monis 4096 Aug 14 14:37 OSLAB14  
drwxr-xr-x 2 monis monis 4096 Aug 7 12:08 OSSP-Practical  
-rwxr-xr-x 1 monis monis 16000 Aug 14 14:38 Program1  
-rw-r--r-- 1 monis monis 353 Aug 14 14:38 Program1.c  
-rw-r--r-- 1 monis monis 353 Aug 14 14:36 Program1.c.save  
-rwxr-xr-x 1 monis monis 16088 Aug 14 14:48 Program2  
-rw-r--r-- 1 monis monis 235 Aug 14 14:45 Program2.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 14:47 Program3  
-rw-r--r-- 1 monis monis 336 Aug 14 14:47 Program3.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 14:52 Program4  
-rw-r--r-- 1 monis monis 418 Aug 14 14:51 Program4.c  
-rwxr-xr-x 1 monis monis 16040 Aug 14 14:54 Program5  
-rw-r--r-- 1 monis monis 376 Aug 14 14:53 Program5.c  
-rwxr-xr-x 1 monis monis 16000 Aug 14 14:56 Program6  
-rw-r--r-- 1 monis monis 288 Aug 14 14:55 Program6.c  
drwxr-xr-x 2 monis monis 4096 Aug 4 10:08 Simple-Commands-Week1  
-rw-r--r-- 1 monis monis 0 Jul 31 03:59 example.txt  
drwxr-xr-x 2 monis monis 4096 Jul 31 04:00 foldername  
-rwxr-xr-x 1 monis monis 15952 Aug 4 03:53 hello  
-rw-r--r-- 1 monis monis 75 Aug 4 03:51 hello.c  
drwxr-xr-x 2 monis monis 4096 Jul 28 04:00 moni  
drwxr-xr-x 2 monis monis 4096 Jul 28 04:09 monis  
-rwxr-xr-x 1 monis monis 16080 Aug 14 09:52 program2  
-rw-r--r-- 1 monis monis 229 Aug 14 09:51 program2.c  
-rwxr-xr-x 1 monis monis 16080 Aug 14 10:06 program3  
-rw-r--r-- 1 monis monis 260 Aug 14 10:06 program3.c  
-rwxr-xr-x 1 monis monis 16040 Aug 14 10:13 program4  
-rw-r--r-- 1 monis monis 163 Aug 14 10:13 program4.c  
-rwxr-xr-x 1 monis monis 16000 Aug 14 10:17 program5  
-rw-r--r-- 1 monis monis 155 Aug 14 10:17 program5.c  
-rwxr-xr-x 1 monis monis 15952 Jul 31 09:36 sample1  
-rw-r--r-- 1 monis monis 77 Jul 31 09:18 sample1.c  
-rw----- 1 monis monis 0 Jul 31 09:17 sample1.c.save  
monis@DESKTOP-C3GM4T6:~$
```